

AUDIO SPECTRUM ANALYSER With Bar-Type LED Display

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An audio spectrum analyser is an electronic instrument or device that measures and displays the amplitude of an input signal as a function of frequency. It provides a visual representation of the frequency spectrum, typically using a row or column of LEDs to indicate the strength of different frequency components present in the audio signal. By separating a complex signal into its individual frequency components, the analyser reveals how energy is distributed across different frequency ranges.

The device presented here is a simple audio spectrum analyser designed to visually represent the frequency content of an audio signal. The input signal is taken directly from the headphone output of a laptop or mobile phone and passed through filters that divide it into different frequency bands, such as bass, mid-range, and treble. Each frequency band drives a group of LEDs that glow according to the signal strength within that band. As the audio changes, the LEDs respond dynamically, producing a visual display that corresponds to the sound being output.

Such analysers are commonly used in audio systems, music players, and sound-responsive lighting displays to provide a visual indication of the audio spectrum. The prototype of the device is shown in Fig. 1.

Circuit and working

Fig. 2 shows the circuit diagram of the audio spectrum analyser. The device is built around op-amp LM358 (IC1), dot/bar display driver LM3914 (IC2), and a few other passive components.

The circuit includes an active band-pass filter designed to pass

frequencies approximately between 200Hz and 500Hz. This enables the device to respond mainly to signals within this frequency range.

The audio signal is taken from the headphone jack of a laptop or mobile phone. Since the input signal is typically stereo, it is first converted to mono using a simple mixer network formed by two 10kΩ resistors (R1 and R2). This mixed signal is then fed into the circuit's filter and processing stages.

To obtain a band-pass response, a high-pass filter is followed by a

low-pass filter. In this circuit, the first stage acts as a high-pass filter implemented using op-amp LM358. This stage blocks very low-frequency components and allows signals above the selected cut-off frequency to pass. The output of this stage is then fed to a low-pass filter, which attenuates higher frequencies. Together, these two stages form an active band-pass filter that allows only a specific range of frequencies to pass through the device.

The audio signal obtained from the headphone jack of a laptop or mobile phone is typically around 400-500mV AC. Since this level is insufficient to directly drive the LED display driver IC, the signal is amplified by LM358 op-amp. The amplifier gain is adjusted so that the output voltage falls within the input operating range of LM3914 display driver. The gain of the amplifier stage

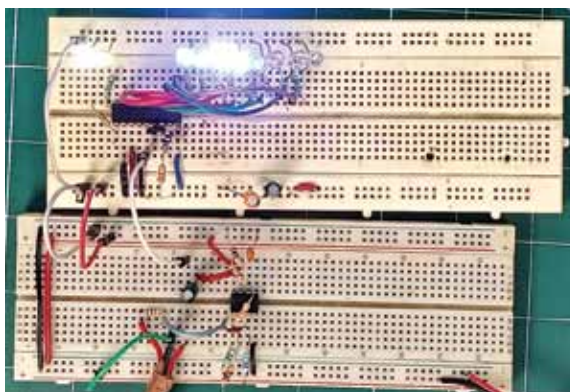


Fig. 1: Author's prototype

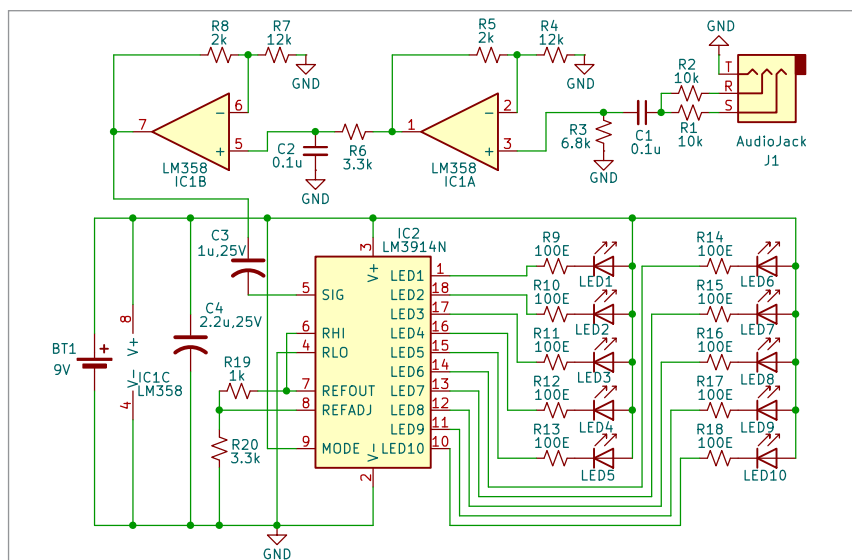


Fig. 2: Circuit diagram